How-To Library

Informatica

Configuring Okta for Snowflake Data Cloud Connector to access Snowflake

© Copyright Informatica LLC 2023. Informatica, Informatica Cloud, and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This article provides instructions on how to authenticate to Snowflake using an external OAuth such as Okta in Snowflake Data Cloud Connector from Cloud Data Integration.

Supported Versions

- Informatica Cloud® Data Integration Connectors

Table of Contents

Overview. 2
Prerequisites. 2
Create a security integration in Snowflake. 2
Configuring Okta using OAuth 2.0 authorization code authentication. 3

Overview

You can integrate your organization external OAuth server such as Okta with Snowflake and use single sign-on (SSO) to connect to Snowflake securely without the need to enter any additional authentication or authorization factor or method.

The external OAuth server helps in generating the OAuth 2.0 access tokens which you can use to connect to the Snowflake warehouse. For more information about external OAuth servers, see External OAuth.

This article describes the steps to configure Okta for Snowflake Data Cloud Connector to connect to Snowflake from Cloud Data Integration. Okta is a standards-compliant OAuth 2.0 authorization server and an OpenID Connect provider. The OAuth 2.0 protocol provides API security through scoped access tokens, and the OpenID Connect provides user authentication and SSO functionality.

Prerequisites

Before you configure Okta in a Snowflake Data Cloud connection, ensure that you have completed the following prerequisites:

- You must have an account in Okta. If the Snowflake user name is John@domain.com, ensure that you use the same user name to create an account in Okta.
- Perform the tasks to integrate Okta with Snowflake. For instructions, see Configure Okta for External OAuth.

Create a security integration in Snowflake

Create a security integration entry for Okta in the Snowflake administration account. An integration is a Snowflake object that provides an interface between Snowflake and your application.

1. Log in as an administrator to Snowflake where you want to create the integration.
2. Enter the security integration details in the following code and run the command:

```
create security integration external_oauth_okta_2
   type = external_oauth
   enabled = true
   external_oauth_type = okta
   external_oauth_issuer = '<OKTA_ISSUER>'
```

```
    external_oauth_jws_keys_url = '<OKTA_JWS_KEY_ENDPOINT>'
    external_oauth_audience_list = ('<SNOWFLAKE_AUDIENCE>')
    external_oauth_token_user_mapping_claim = 'sub'
    external_oauth_snowflake_user_mapping_attribute = 'login_name';
```

For example,

```
create security integration external_oauth_okta_2
    type = external_oauth
    enabled = true
    external_oauth_type = okta
    external_oauth_issuer = 'https://200000000.okta.com/oauth2/aus4000/.well-known/oauth-
authorization-server'
    external_oauth_jws_keys_url = 'https://200000000.okta.com/oauth2/aus4000/v1/keys'
    external_oauth_audience_list = ('https://infapartner.us-
central1.gcp.snowflakecomputing.com')
    external_oauth_token_user_mapping_claim = 'sub'
    external_oauth_snowflake_user_mapping_attribute = 'login_name';
```

Configuring Okta using OAuth 2.0 authorization code authentication

After creating the security integration, create a Snowflake Data Cloud connection in Cloud Data Integration. In the connection, select the Authorization Code authentication type to connect to Snowflake and enter the configured Okta details.

After you enter the details, you can authenticate using Okta SSO. The token along with the refresh token gets populated in the connection parameters. You can then test the connection to check if the connection is successful to Snowflake.

The following table describes the Snowflake Data Cloud connection properties to configure Okta in an OAuth 2.0 - AuthorizationCode type connection:

Property	Description
Connection Name	Name of the connection.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks.
Authentication	The authentication method that Snowflake Data Cloud Connector must use to log in to Snowflake. Select **AuthorizationCode**.
Account	The name of the Snowflake account.
Warehouse	The Snowflake warehouse name.
Authorization URL	The Snowflake server endpoint that is used to authorize the user request. The authorization URL is https://{Okta_Domain}/oauth2/{authorization Server ID}/v1/authorize. To get the authorization URL from Snowflake, navigate to Security>API>Consider Issuer URI/v1/authorize.
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code for an access token. The access token URL is Issuer_URI/v1/token. For example, https://<abc>.snowflakecomputing.com/oauth/token-request **Note:** If the account name contains underscores, use the alias name.

Property	Description
Client ID	Client ID of Okta.
Client Secret	Client secret of Okta.
Scope	Determines the access control if the endpoint has defined custom scopes. Enter space-separated scope attributes. For example, specify `session:role:<rolename> openid offline_access` as the scope to override the value of the default user role. The value is the role assigned in Security Integration. **Note:** You need to add *openid offline_access* so that the token gets refreshed after expiry.
Access Token Parameters	Additional parameters to use with the access token URL.
Authorization Code Parameters	Additional parameters to use with the authorization token URL.
Access Token	The access token value. Enter the populated access token value, or click **Generate Access Token** to populate the access token value.
Generate Token	Generates the access token and refresh token based on the OAuth attributes you specified.
Refresh Token	The refresh token value. Enter the populated refresh token value, or click **Generate Access Token** to populate the refresh token value.

Authors

Divya Pai

Dimple Rai